HEIG-VD HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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Formal verification
Verification of an AXI-Stream upscaler
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Context

We want to verify an AXI-Stream module that combines several narrow input transfers into one wider output transfer. The reference configuration has an input width of 32 bits and an output width of 128 bits, so four input words are packed into each output word.
The module interface is :

```
entity axis_upscaler is
    generic (
        DATA_IN_WIDTH  : integer := 32;
        DATA_OUT_WIDTH : integer := 128
    );
    port (
        clk            : in  std_logic;
        rst            : in  std_logic;
        s_axis_tdata   : in  std_logic_vector(DATA_IN_WIDTH-1 downto 0);
        s_axis_tvalid  : in  std_logic;
        s_axis_tready  : out std_logic;
        s_axis_tlast   : in  std_logic;
        m_axis_tdata   : out std_logic_vector(DATA_OUT_WIDTH-1 downto 0);
        m_axis_tvalid  : out std_logic;
        m_axis_tready  : in  std_logic;
        m_axis_tlast   : out std_logic
    );
end entity axis_upscaler;
```

A transfer is accepted on a rising edge of `clk` when both `tvalid` and `tready` are high. On the input, these signals are `s_axis_tvalid` and `s_axis_tready`; on the output, they are `m_axis_tvalid` and `m_axis_tready`. The source must keep its valid signal and payload stable until an input transfer is accepted. The reset `rst` is active high and asynchronous.

Required behavior

Use the default configuration first. Let

$$F = \frac{\texttt{DATA_OUT_WIDTH}}{\texttt{DATA_IN_WIDTH}} = 4.$$

The widths must be positive, and the output width must be an integer multiple of the input width.

1. Accumulate accepted input words in order. The first accepted word occupies the least-significant `DATA_IN_WIDTH` bits of the output word, the second occupies the next slice, and so on. No accepted input word may be lost, duplicated, or reordered.
2. Once `F` input words have been accepted, make the assembled output word valid. An output transfer occurs only when `m_axis_tvalid` and `m_axis_tready` are both high.
3. `s_axis_tlast` marks the final accepted input word of a frame. The corresponding assembled output word must carry `m_axis_tlast`. If the frame ends before `F` words have been collected, zero-fill the unused higher-order slices of that output word.
4. While `m_axis_tvalid` is high and `m_axis_tready` is low, keep `m_axis_tvalid`, `m_axis_tdata`, and `m_axis_tlast` stable until the output transfer occurs.

For example, if the input words are `A`, `B`, `C`, and `D`, the output slices from least to most significant contain `A`, `B`, `C`, and `D`. If a frame ends after `A` and `B`, the output contains `A`, `B`, zero, and zero, with `m_axis_tlast` asserted.

Verification task

Write PSL properties in the vunit bound to `axis_upscaler`.
Try to verify at least :

- input and output ready/valid behavior, including stability under backpressure ;
- propagation of `tlast` to the matching output transfer, including when a frame ends early ;
- data placement, ordering, and zero padding ;

Error injection

A generic parameter, `ERRNO`, can be used to inject errors into the design. Its behavior is as follows :

1. When `ERRNO` is 0, the design behaves correctly.
2. When `ERRNO` is between 1 and 6, inclusive, the design contains an error.